

Frank Chen

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EDUCATION

Carnegie Mellon University, Pittsburgh, PA Aug 2026 – Dec 2027 (Expected)
M.S. in Computational Data Science, School of Computer Science QPA: 4.33/4.33, Summer 2026
The Chinese University of Hong Kong, Shenzhen, China Sept 2022 – Jul 2026
B.E. in Computer Science and Engineering, Artificial Intelligence Stream GPA: 3.85/4.00, Top 3%
Coursework: Deep Learning, Reinforcement Learning, Probability & Statistics, Algorithms, Advanced Algebra, Computer Arch
Research Experience: UCLA Visiting Student, Statistics (2025); NUS Exchange Student & Research Assistant (2024)
Honors & Teaching: Academic Performance Scholarship (Top 3%, ×3); Dean's List (×3); Student Teaching Fellow Award

INDUSTRY EXPERIENCE

Machine Learning Engineer Intern | Alibaba Inc. May 2026 – Aug 2026

- Owned a **long-horizon evolving LLM Agent for e-commerce** product taxonomy and attribute (Category–Property–Value) **schema generation**, upgrading workflows with **modular planning, self-evolution, rule learning, and targeted repair**.
- Built a reusable **Agent evaluation benchmark** with 15 metrics and 500+ production categories, spanning common long-tail cases, covering accuracy, convergence, latency, human intervention, and robustness.
- Iterated four production-ready Agent versions, improving first-pass quality from 19% to **65%**, successful repair rate from 8% to **75%**, and correct-content retention from 93% to **98%**, with comparable latency.
- Improved Agent reliability by automatically learning reusable business rules, applying deterministic checks to enforce them, and distilling skills to call MCP to improve accuracy; validated quality with both LLM-based evaluation and data distributions.
- LLM post-training.** Built an 8K-example SFT dataset and fine-tuned **Qwen3-14B** with 4-bit **QLoRA**, including data deduplication, hard-case synthesis, FP16 training, and FlashAttention on 2x NVIDIA L20 GPUs.

Software Engineer Intern | Microsoft Jan 2026 – Mar 2026

- Built components of a **repository-scale Coding Agent** for bug localization, code modification/review, and automated pull-request generation; implemented an end-to-end workflow from bug detection to PR submission.
- Optimized agent **execution, memory/log management, and cross-module observability** with OpenTelemetry, enabling task-level tracing, resource profiling, and debugging across long-running agent workflows.
- Integrated **MCP-based tool interfaces**, optimized prompts and agent–tool interaction, and evaluated reliability across recommendation-system and frontend bug-fixing scenarios.

Technical Co-founder | Shenzhen Siku Intelligent Information Co., Ltd. May 2025 – Aug 2025

- Co-developed **Shuta AI**, an agentic AI learning platform; implemented RAG with LangChain, prompt/context optimization, API benchmarking, and FastAPI backend integration.
- Improved long-document ingestion with MinerU and delivered an end-to-end AI app for multilingual course access and Q&A.

COMPETITIONS

National Olympiad in Informatics of China (NOI), Bronze Medal, 2021
Asia and Pacific Informatics Olympiad (APIO), Silver Medal, 2020

PUBLICATIONS

- Fast-scBatch** (Chen, F. et al., 2024), *ACM-BCB 2024*. **First author**. doi.org/10.1145/3698587.3701383
- MediHerb** (Liu, X.* , Chen, F.* et al., 2025), *Computational Biomedicine*. **Co-first author**. doi.org/10.70401/cbm.2025.0003

RESEARCH EXPERIENCE

Sandboxed Execution of High-Risk OpenClaw Medical Skills | CUHK-Shenzhen Feb 2026 – Mar 2026

- Developed and open-sourced biomedical Agent skills/evaluation workflows; reached **1.2K+ GitHub stars in 7 days**; designed risk-tiered controls and sandboxed execution for privacy-sensitive medical tasks.

Attention-based Multimodal Model for TCM Herb–Disease Inference | Independent Study Jul 2025 – Oct 2025

- Developed a multimodal attention model integrating molecular fingerprints, physicochemical features, molecular graphs, SMILES, and biomedical text for multi-label inference; published in *Computational Biomedicine*.

Statistical Methods for Single-cell Cell Topology | UCLA Sept 2024 – Jul 2025

- Applied statistical hypothesis testing and empirical simulation to infer cellular topology and potential motility from single-cell RNA-seq data, strengthening experience in statistical inference and noisy biological data.

Fast-scBatch: Neural Network-driven Batch Effect Correction | CUHK-Shenzhen Jul 2023 – May 2024

- Developed neural-network methods for batch-effect correction in single-cell RNA-seq data, reducing computational cost with dimensionality reduction and random sampling; published at *ACM-BCB 2024*.

TECHNICAL SKILLS

LLM/AI: LLM Agents, Agent Evaluation, Prompt Engineering, RAG, SFT, LoRA, Fine-tuning, MCP, vLLm, DeepSpeed
Programming/Systems: Python, R, SQL, C/C++, Java; PyTorch, LangGraph, FastAPI, OpenTelemetry, Git, Linux, Docker